

Unit - 1 : Database System Concepts (12 Marks)

Unit Outcomes :-

- State the importance of DBMS over file processing in the given situation.
- Describe the overall structure of the given DBMS.
- Identify relevant database model in the given situation.
- Draw the E-R diagram of the given database and identify relationship between the entities.

1.1 Concept of Data, Database & DBMS :-

Data :- Data are raw or isolated facts from which the required information is produced.

Examples of Data :-

- In Employer's mind - Emp_id, Emp_name, Emp_dept, DOB etc.
- In Sales person's view -
Customer_name, customer_account, address, city, ph_no etc.

Information :- Information is the result of processing raw data to reveal its meaning.

Database :- A database is defined as a collection of logically related data, stored together, that is designed to meet the information needs of an organization.

A Database consists of the following components :

- Data items
- Relationships
- Constraints
- Schema.

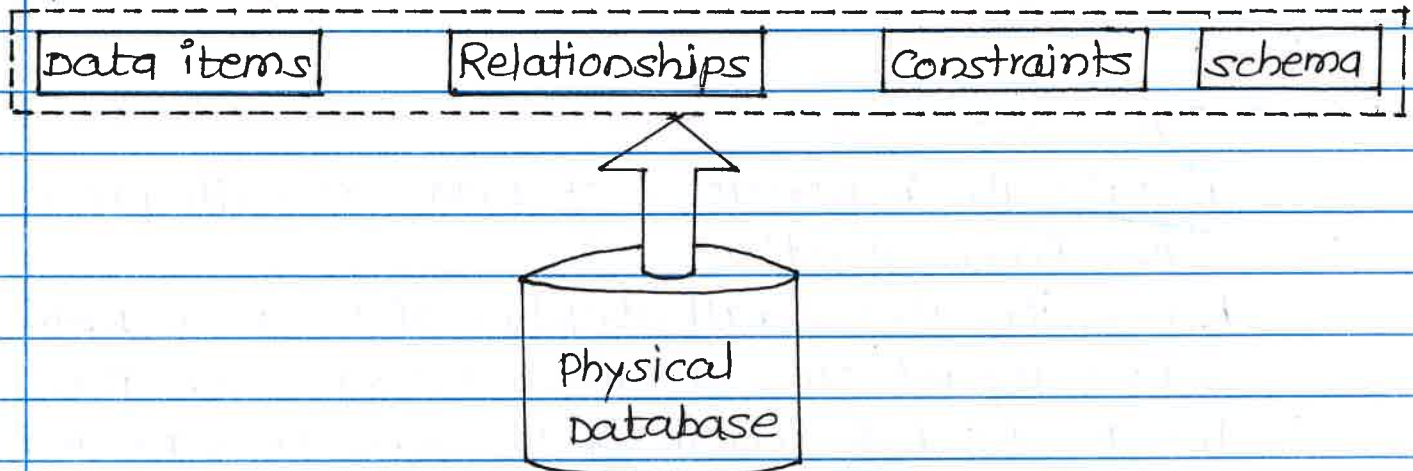


fig. Components of a Database.

Database Management System (DBMS) :-

Defⁿ:- A Database Management System is a collection of interrelated data and a set of programs to access those data.

Objective:- The objective of DBMS is to provide a convenient and effective method of defining, storing and retrieving the information contained in the database.

- Database systems are designed to manage large amount of information and providing mechanisms for manipulation of information.
- In addition, the database systems must ensure the safety of the information stored despite system crashes or unauthorized access.

Advantages of DBMS over file processing systems :-

1. Improved data sharing :-

The DBMS helps to create an environment in which end users have better access to more and better managed data. Such access makes it possible end users to respond quickly to changes in their environment.

2. Better Data Integration :-

Wider access to well-managed data promotes an integrated view of the organization's operations and a clearer view of the big picture. It becomes easier to see how actions in one segment of the company affects other segments.

3. Minimized Data Inconsistency :-

Data inconsistency exists when different versions of the same data appear in different places. The probability of data inconsistency is reduced in a properly designed database.

4. Improved data access :-

The DBMS makes it possible to produce quick answers to the queries.

5. Improved decision making :-

Better managed data and improved data access make it possible to generate better quality information, on which better decisions are based.

6. Increased end-user productivity :-

The availability of data, combined with the tools that transform data into usable information, empowers end users to make quick informed decisions that can make the difference between success and failure.

Applications of Database systems :-

Databases are widely used in -

1. Banking :- For storing customer information, accounts, loans and banking transactions etc.
2. Airlines :- for reservations and schedule information.
3. Universities :- for student information, course registration, grades/results etc.
4. Telecommunication :- for keeping records of calls made, generating monthly bills, maintaining balances on prepaid calling cards and storing information about communication networks etc.
5. Railways
6. Finance
7. Sales
8. Payroll systems
9. Manufacturing
10. Human Resources.

1.3 Data Abstraction :-

One of the major purpose of database systems is to provide users an abstract view of data i.e. the system hides certain details of how the data is stored and maintained.

For the system to be usable, it must retrieve data efficiently. The need for efficiency has led designers to use complex data structures to represent data in the database. Since many database system's users are not computer-trained, developers hide the complexity from users through several levels of abstraction to simplify user's interaction with the system.

There are three levels of data abstraction

- i) Physical Level or Internal level
- ii) Logical Level or conceptual level
- iii) View Level or External level.

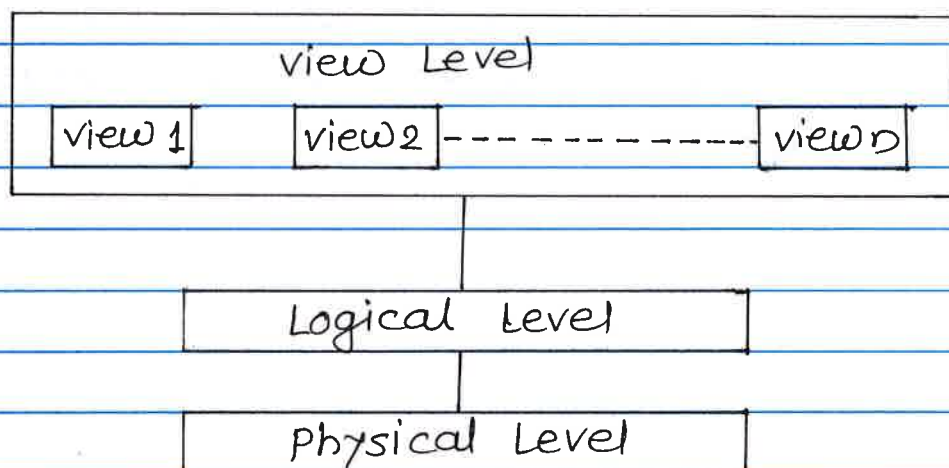


fig. Three Levels of data abstraction.

Physical Level :- This lowest level of abstraction describes how the data is actually stored. The physical level describes complex low level data structures in detail.

Logical Level :- The next higher level of abstraction describes what data are actually stored in the database and what relationship exist among those data.

The logical level thus describes entire database in terms of small number of simple structures. Database administrators, who decides what information to keep in the database use logical level of abstraction.

View level :- The highest level of abstraction describes only part of the entire database. Even though, the Logical level uses simple structures, complexity remains because of the variety of information stored in the large database. Many users of the database system do not need all this information; instead, they need to access only part of the database. The view level of abstraction exists to simplify their interaction with the system. The system may provide many views for the same database.

Instances and Schema :-

Instances :- The collection of information stored in the database at a particular moment is called an instance of the database.

Schema :- The overall design of the database is called the database schema.

The database schema can be partitioned according to the levels of abstraction:

1. Physical or Internal Schema :- The physical schema, describes the physical storage structure of the database.

This level is concerned with the following activities:

- i) storage space allocation for data and storage
- ii) Record descriptions for storage with stored size for data items
- iii) Record placement
- iv) Data compression and data encryption techniques.

This schema uses a physical data model and describes the complete details of data storage and access path for the database.

2. Logical or Conceptual Schema :- The logical schema describes the structure of the whole database for a community of users. The conceptual schema describes the entities, data types, relationships, user operations and constraints and hides the details of physical storage structure.

The conceptual level is concerned with following activities:

- i) All entities, their attributes and their relationships.
- ii) Constraints on the data
- iii) Semantic information about the data.

iv) security information

v) checks to retain data consistency and integrity.

3. External schema or Subschema:- External schema describes the database schema at the view level that describe different views of the database. Each external schema describes the part of the database that a particular user group is interested in and hides the rest of database from that user group.

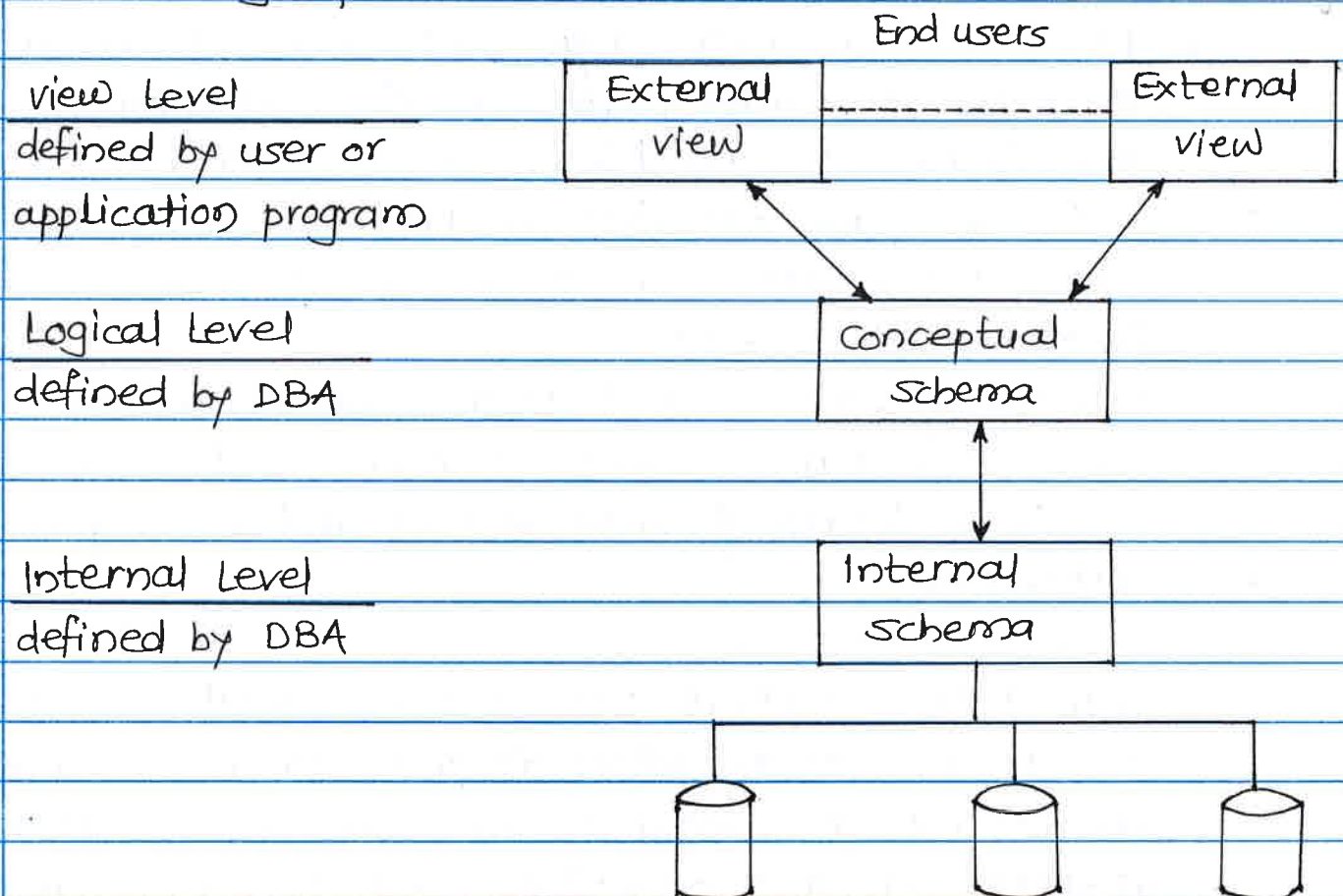


fig. Three schema architecture.

1.4: Overall structure of DBMS :-

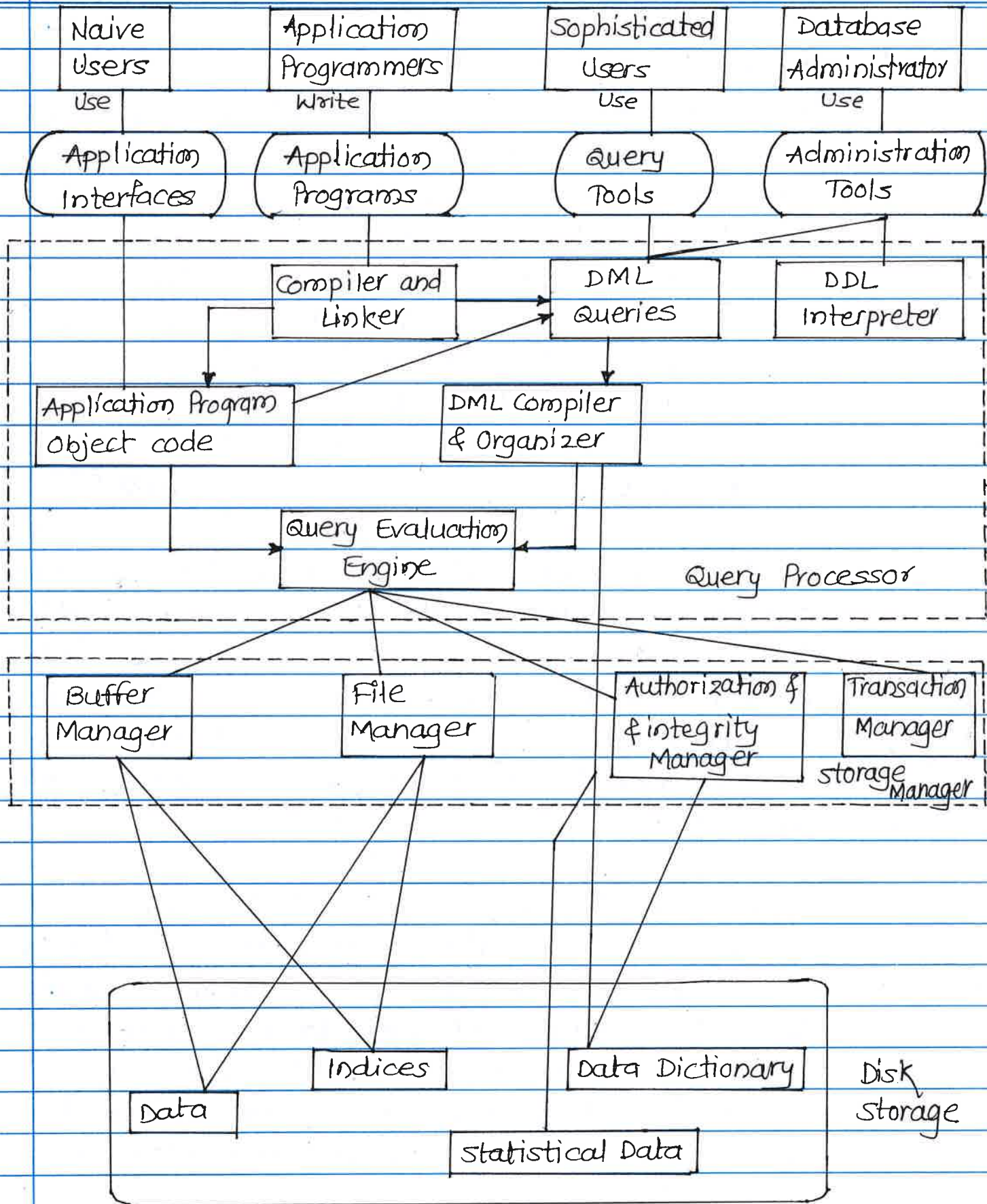


fig. Overall structure of DBMS.

Database Users :-

There are four different types of database system users.

1. Naïve Users :- These are unsophisticated users who interact with the system by invoking one of the application programs that have been written previously.
 - Users who need not be aware of the presence of the database systems or any other system supporting their usage are considered naïve users.
 - The operations that can be performed by this class of users are very limited and affect a precise portion of the database.
2. Application Programmers :- Application programmers are computer professionals who write application programs. Application programmers can choose from many tools to develop user interfaces. For ex. Rapid Application Development (RAD) tools are the tools that enable an application programmer to construct forms & reports without writing a program.
3. Sophisticated Users :- These users interact with the system without writing application programs. Instead, they form their requests in a database query language. They submit each such query to a query processor, whose function is to break down DML statements into instructions that the storage manager understands. Analysts who submit queries to explore data in the database fall in this category.

4. Specialized Users :- These are sophisticated users who write sophisticated database applications that do not fit into the traditional data processing frameworks. Among these applications are Computer Aided Design systems, Knowledge based and expert systems etc.

Database Administrators :-

A person who has central control of both the data and the programs that access those data over the system is called a database administrator.

Functions/Roles/Responsibilities of DBA :-

Database administrators are the users who are most familiar with the database and are responsible for creating, modifying and maintaining its three levels. Functions of DBA include:

1. Defining logical schema & physical schema
2. Defining security & integrity constraints.
3. Defining backup and recovery procedures.
4. Storage structure and access method definition.
5. Monitoring performance and responding to changing requirements.
6. Granting of authorizations for data access.
7. Schema and physical organization modification.
8. Ensuring that enough free disk space is available for normal operations and upgrading disk spaces as required.

Query Processor :-

1. DML Compiler :- It translates DML statements into a query language into low level instruction that query evaluation engine understands.
2. DML Compiler :- It converts DML statements embedded in an application programs to normal procedure calls in the host language. The precompiler interact with DML compiler to generate the appropriate code.
3. DDL Interpreter :- It interprets DDL statements & records the definitions in the data dictionary.
4. Query Evaluation Engine :- It executes low-level instructions generated by the DML compiler.

Storage Manager :-

1. Buffer Manager :- It is responsible for fetching data from disk storage into main memory and deciding what data to fetch in main memory.
2. File Manager :- It manages the allocation of space on disk storage and data structures used to represent information stored on disk.
3. Authorization & integrity Manager :- It tests for the satisfaction of integrity constraints and checks the authority of users to access data.
4. Transaction Manager :- It ensures that database remains in a consistent state despite system failures & current transaction executions without conflicting.

There are following data structures required as part of physical system implementation.

- i) Data File : It stores the data itself.
- ii) Data Dictionary : It stores metadata about the structure of the database, in particular, the schema of the database.
- iii) Indices : It provides fast access to the data items that hold particular values.
- iv) Statistical Data : It stores statistical information about the data in the database.

1.5. Data Modeling :-

Data Models :- Definition :- A collection of concepts that can be used to describe the structure of the database is called a data model.

The structure of a database means datatypes, constraints and relationships that should hold on the data.

Record based Logical Models :-

1. Relational Model :-

The relational model uses a collection of tables to represent both data and relationships among those data.

A table is a collection of rows and columns. Each column has an unique name. Each row in the table represents collection of related data values. In the relational model, a row is called a tuple, a column header is called an attribute and the table is called a relation.

Advantages :-

1. Simplicity :- A relational data model is even simpler than the hierarchical and network models. It frees designers from the actual physical data storage details, thereby allowing them to concentrate on the logical view of the database.
2. Structural Independence :- The relational data model does not depend on the navigational data access system. changes in the database structure do not affect the data access.
3. Ease of design, implementation, maintenance and users.
4. Flexible and powerful query compatibility.

Disadvantages:

1. Hardware overheads :- The relational data models need more powerful computing hardware and data storage devices to perform RDBMS - assigned tasks.
2. Easy to design capability leading to bad design.

2. Network Data Model :-

Data in the network model represent the collection of records and relationships among the data are represented by links, which can be viewed as pointers.

The records in the database are organized as collection of arbitrary graphs.

Advantages:

- i) Simplicity :- Network model is simple & easy to design.
- ii) Superior data access :- Data access and flexibility is superior than hierarchical data model.
- iii) facilitates more relationship types :- The network model

facilitates in handling of one-to-many (1:n) and many-to-many (m:n) relationships, which helps in modeling real life situations.

i) Database integrity :- Network model enforces database integrity and does not allow a member to exist without an owner.

ii) Data independence :- Network data model provides data independence by at least partially isolating the programs from complex physical storage details.

Disadvantages :-

i) System complexity :- Since network model provides navigational access mechanism to the data in which data are accessed one record at a time. This mechanism makes the system implementation very complex.

ii) Absence of structural independence :- It is difficult to make changes in a network database.

iii) Not user-friendly :- The network data model is not a design for user-friendly system and is a highly skill-oriented system.

iv) The use of pointers leads to a complex structure, which make mapping of related data very difficult.

3. Hierarchical Data Model :-

The hierarchical model provides two main structuring concepts -

i) Records

ii) Parent-child relationships.

Records - A record is a collection of field values that provide the information on an entity or a relationship instance.

Parent-child Relationship :- A parent-child relationship type is a 1:N relationship between two record types.

- The record type on the 1-side is called parent record and the one on N-side is called child record.

An instance of parent-child relationship consists of one record of parent type and a number of child records.

Advantages :-

- i) Simplicity :- The relationship between various layers is logically simple and design of a hierarchical database is simple.
- ii) Data sharing :- Because all data are held in a common database, data sharing becomes practical.
- iii) Data Security :- It was the first database model that offered data security that is provided and enforced by the DBMS.
- iv) Data integrity :- The parent-child relationship, there is always link between parent segment & its child segments under it.
- v) Data independence :- The DBMS creates an environment in which data independence can be maintained.

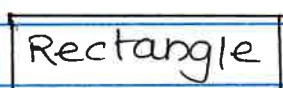
Disadvantages :-

- i) Implementation Complexity :- Although hierarchical database is conceptually simple, easy to design and no data independence problem, it is quite complex to implement.
- ii) Implementation Limitation :- Many of the common relationships do not confirm to the one-to-many relationship format required by hierarchical data model.
- iii) Inflexibility :- A hierarchical database lacks flexibility. The changes in the new relations or segments often yield very complex system management tasks.
- iv) Lack of structural independence.
- v) Application Programming complexity.

1.6. Data Modeling using E-R Model :-

Entity Relationship Model :-

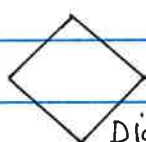
The E-R Data Model is based on a perception of a real world that consists of a collection of basic objects called entities and relationships among these objects. The overall logical structure of a database can be represented graphically by E-R diagram. The E-R diagram is build up from following components:

 Rectangle

→ Represents entity sets.

 Ellipse

→ Represent Attributes



Diamond

→ Represents relationships among entity sets

Lines

→ Links attributes to entity sets and entity sets to relationship sets.

double ellipse

→ Represents Multivalued Attributes.

dashed ellipse

→ Represents Derived Attributes.

double
Rectangle

→ Represents Weak Entity sets.

Double Line

→ Represents total participation of an entity in a relationship set.

Advantages :-

- i) straight forward relational representation.
- ii) Easy conversion for E-R to other model.
- iii) Graphical representation for better understanding.

Disadvantages :-

- i) No industry standard for notation
- ii) Popular for high level design.

Attributes :- Each entity is represented by a set of properties called as attributes.

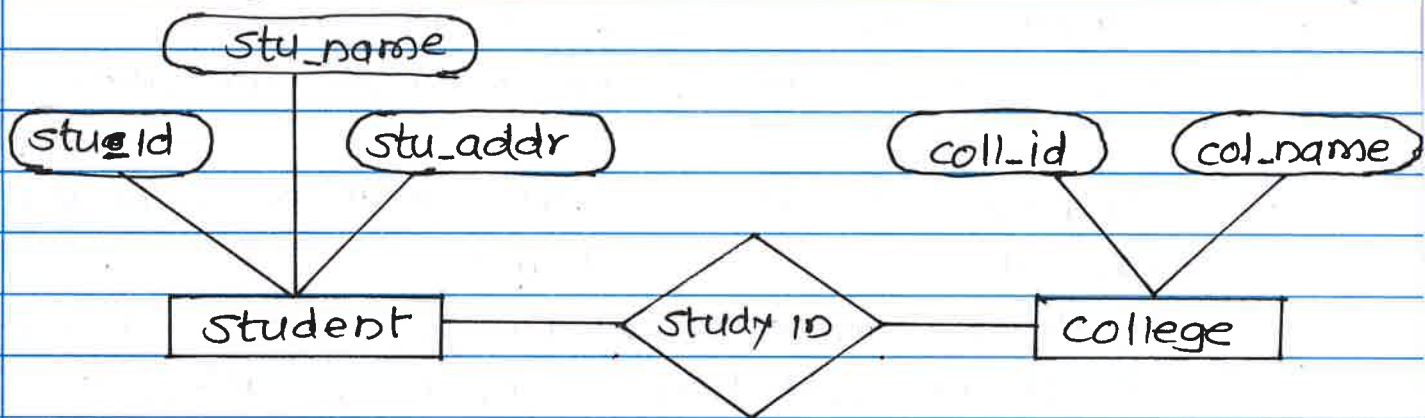
Key Attributes :- An entity type usually has an attribute whose values are distinct for each individual entity in the collection. Such an attribute is called a key attribute and its values can be used to identify each entity uniquely.

Types of attributes :-

- 1) Simple Attributes :- Attributes that cannot be divided into subparts are called simple attributes.
Ex: Roll_no is an example of simple attributes.
- 2) Composite Attributes :- Attributes that can be divided into subparts are called composite attributes.
Ex: Name ~~is~~ ~~an~~ ~~example~~ of composite attribute, because it can be divided into first_name, middle_name & last_name.
- 3) Single-valued Attributes :- Single-valued attributes have a single value for a particular entity.
Ex: loan_number attribute for a specific loan entity refers to only one value. Such attributes are single-valued attributes.
- 4) Multi-valued Attributes :- An attribute which may have a set of values is called multi-valued attribute.
Ex: An employee set with the attribute phone_no. is an example of multi-valued attribute because an employee may have one or many phone numbers.

- 5) Derived Attributes :- The value for this type of attribute can be derived from the values of other related attributes.
Ex: Let us consider an entity set is having attribute DOB, we can derive age from DOB. Here, age is a derived attribute.
- The value of the derived attribute is not stored, but it is computed whenever required.

E-R Diagrams :-



E-R Diagram specifying relationship between student and college.